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DETAIL SPECIFICATION
MANUALS, TECHNICAL - WORK UNIT CODE

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AMSC 32776 **AREA TMSS**

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This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification prescribes the requirements for the development and preparation of Work Unit Code (WUC) Technical Manuals (TMs) for Air Force equipment. Appendix B provides information for utilization of markup language tools for the digital preparation of technical data to be delivered to the Government.

1.2 Illustrations in this specification. The illustrations appearing in this specification are used only as examples. If there is any conflict between the text and illustrations of this document, the text applies.

2 APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-38784 - General Style and Format Requirements for Technical Manuals

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AIR FORCE TECHNICAL MANUALS

TO 00-20-2 - Maintenance Data Documentation

(Copies of this document required by users with ".mil" government web address access are available online at https://www.my.af.mil/etims/ETIMS/index.jsp. Copies of documents required by contractors in connection with specific procurement functions should be obtained from the acquiring activity or as directed by the contracting officer.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3 REQUIREMENTS

3.1 Development and preparation. The general manner of development and preparation of WUC manuals shall be in accordance with the requirements of MIL-STD-38784, except as otherwise specified herein (see 6.6).

3.1.1 Manual size. Preliminary manuals shall be 8½ by 11-inches (with the same printing area size as 5 by 8-inch manuals). Formal manuals shall be prepared in 5 by 8-inch size, except training and Support Equipment (SE) manuals which shall be prepared in 8½ by 11-inch size.

3.1.2 Capitalization. For WUC entries, each line entry with a zero as the fifth character of the WUC shall be in upper-case. Each line entry with other than a zero as the fifth character of the WUC shall be in lower-case, with the first letter of each word in upper-case (see figure 1). For support general codes, the title of each code shall be in

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upper-case. Tasks within that code shall be in lower-case, with the first letter of each word in upper-case. Descriptive text shall follow normal sentence capitalization.

3.1.3 Column arrangement. There shall be a blank line between WUC upper-case line entries (see 3.1.2) and the preceding line entry (excluding support general codes).

Paper output: 5 by 8 inch manuals shall be single column and 8½ by 11-inch manuals shall be double column.

3.1.4 Page numbering. The TOC, introduction, type maintenance, action taken, when discovered, and how malfunctioned codes pages shall be numbered with an upper case roman numeral for each category (when discovered, how malfunctioned codes avionic/electrical/computer [alphabetic sequence], etc.) and the page sequence within that category (see figure 2 and 6.12). Pages containing WUCs shall be numbered by the WUC system (or homogeneous group see 6.4.3) number and consecutive page number (see 6.12). Pages containing System Subsystem Reference Designation Indicator (SSRDI) numbers shall be numbered by the system/subsystem number and consecutive page numbers (see 6.12). Support General Codes 01000, 02000, 05000, 06000, 07000, and 09000 shall be listed on pages prefixed 01- (see TO 00-20-2, appendix I). Codes 03XXX and 04XXX shall appear on pages prefixed 03- and 04-, respectively (see TO 00-20-2, appendix I).

3.1.5 Page entries. Paper output: When a change occurs in the system number, i.e., 20-, 21-, 22-, the listing for each system shall begin on a new right-hand page.

PDF output: When a change occurs in the system number, i.e., 20-, 21-, 22-, the listing for each system shall begin on a new page.

3.1.6 Standardization. When similar systems and components are used in two different series of the same mission design aircraft, such as the F-15 A/B and the F-15 C/D, the first three characters of the WUC shall be the same in both manuals. To the maximum extent practicable, the fourth and fifth characters of the WUC should also be the same.

3.1.7 Test program. Unless otherwise specified by the acquiring activity (see 6.2.b), when the contract contains a requirement for the collection of maintenance management type data during a test program (see 6.7), the WUC manual shall be completed prior to the start of the test program. The contractor shall be responsible for making any necessary changes to the manual resulting from the test program, including the addition of new end items.

3.2 Work unit code manual. Appendix B provides the markup language tools for the electronic delivery of this manual. The content and arrangement of the WUC manual shall be as specified below. If required by the acquiring activity, SE shall be included (see 6.2.c and 6.11). The manual shall not contain chapters or sections.

a. Front matter (see 3.2.1).

b. Type maintenance codes (see 3.2.2.1).

c. Action taken codes (see 3.2.2.2).

d. When discovered codes (munitions and SE codes, when applicable, shall be in separate lists) (see 3.2.2.3).

e. How malfunctioned codes alphabetical listing-avionics/electrical/computer (see 3.2.2.4).

f. How malfunctioned codes alphabetical listing-physical/mechanical (see 3.2.2.4).

g. How malfunctioned codes alphabetical listing-engine related (see 3.2.2.4).

h. How malfunctioned codes numerical listing-avionics/electrical/computer (see 3.2.2.4).

i. How malfunctioned codes numerical listing-physical/mechanical (see 3.2.2.4).

j. How malfunctioned codes numerical listing-engine related (see 3.2.2.4).

k. Support general codes (except 03000 and 04000) (see 3.2.3).

l. 03000 Support general codes (see 3.2.3).

m. 04000 Support general codes (see 3.2.3).

n. Work unit code - noun - system/subsystem/reference designation index (see 3.2.4).

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o. System/subsystem/reference designation index - noun - work unit code (see 3.2.4).

p. Work unit codes - noun - equipment identification (SE only) (see 3.2.4).

3.2.1 Front matter. Front matter shall be prepared in accordance with the requirements of MIL-STD-38784, except as otherwise specified herein.

3.2.1.1 Table of contents (TOC). The WUC listing shall consist of the first two characters of the support general or equipment end item (except for homogeneously grouped items) (see 6.4.3) and the title or narrative description of each code listed (see figure 2). Homogeneously grouped items shall be listed under the homogeneous group title in alphabetic sequence by end item nomenclature, with the complete end item WUC contained in the code column of the TOC.

3.2.1.2 Introduction. In addition to the requirements of MIL-STD-38784, the introduction shall contain a brief description of the codes and their use.

3.2.2 Type maintenance, action taken, when discovered, and how malfunctioned codes. These codes are defined and listed in TO 00-20-2 (see 6.5 and 6.8).

3.2.2.1 Type Maintenance Code (TMC). This list shall include only those codes that are applicable to the equipment covered by the WUC manual (see 6.4.8). The following note shall be included in all WUC manuals following the TMCs:

NOTE

Refer to TO 00-20-2, appendix F for off-equipment and shop type maintenance codes.

3.2.2.2 Action Taken Code (ATC). The complete list of action taken codes shall be included in the manual (see 6.4.1).

3.2.2.3 When Discovered Code (WDC). Only the when discovered codes and definitions, for the equipment to which the WUC manual is applicable, shall be included in the manual (see 6.4.10).

3.2.2.4 How Malfunctioned Code (HMC). The how malfunctioned code shall be listed in numeric sequence and shall contain only those codes that are applicable to the specific equipment contained in the WUC manual (see 6.4.4).

3.2.2.4.1 Alphabetic and numeric sequence. The alphabetic and numeric sequence shall be further divided into the following four categories:

a. Avionics/Electrical/Computer.

b. Physical/Mechanical.

c. Engine Related.

d. No Defect.

3.2.2.4.2 Engine related HMCs. Engine related HMCs are restricted and shall be further divided into six categories:

a. Observed or Recorded Operational Conditions.

b. Identified Components.

c. Condition Monitoring.

d. Chance Occurrences.

e. Managerial Decision.

f. Cause for Removal

These codes shall be used for engine related items (see 6.4.4.1).

3.2.2.4.3 Restricted uses of HMC. The following note shall immediately precede the alphabetical listing of HMC:

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NOTE

Technicians must use a How Malfunctioned Code (HMC) from the engine related codes when documenting the removal of engines or engine components that are AFMAN 20-116 reportable. For all other maintenance actions, on any type of equipment, there are no restrictions on the use of codes by category. If a code is appropriate for an observed malfunction, it may be used regardless of the category in which it is listed. Avionics shops may use engine codes. Engine shops may use avionics codes. Tire shops may use physical/mechanical, avionic, or engine codes, etc. Categories of codes were created primarily for ease in finding a specific code.

3.2.3 Support general codes. (See 6.4.7).

3.2.4 Work unit code. (See 6.4.11). Construction and application of WUCs shall be in accordance with appendix A.

3.2.4.1 Used with, but not part of. Work Unit Codes for these items shall be included (see 6.4.9).

3.2.4.2 Nomenclature. Each WUC in the equipment identification listing shall reflect the information required to properly identify all levels of assembly and shall include the basic noun for each individual item that is coded. If an official military type nomenclature has been assigned (per Illustrated Parts Breakdown (IPB) TM, equipment drawings, etc.), it shall be used for identification. Part numbers may be used to identify items if no official nomenclature exists. The basic part number only shall be used (i.e., 47A102212-XX rather than 47A102212-05). Abbreviations may be used when specified by the acquiring activity (see 6.2.d).

3.2.4.3 Time Change Item (TCI), configured article, serially controlled, and warranty tracked items. (See 6.9). Items shall be denoted, between the WUC and its definition (nomenclature), as listed below:

C	Configured article
*****	Serially tracked item
T	Time change item
W	Warranty tracked

These symbols may also be used in combinations, except "*" shall not be used with "C" or "W". Additionally, time change items are further identified by the abbreviation (TCI) following the nomenclature. An explanation of all symbols used shall be provided in the manual introduction.

EXAMPLE:

Time change (T)

99ABC	T	Ignitor (TCI)
99ABB	T	Cable, Igniter (TCI)

Configured article (C)

99DEF	C	Stator

Serially controlled (*)

99TUV	*	Housing
12E00	*	EJECTION SEAT, ACES II FWD & AFT

Time change items are further identified by the abbreviation (TCI) following the definition.

12EGA	T	Chute, Personnel 28 Ft (TCI)

Warranty tracked (W)

99XYZ	W	Amplifier

3.2.4.4 End item identification. Each end item of SE shall be identified by an "@" (at) symbol between the WUC and its nomenclature (see 6.4.2).

EXAMPLE:

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SA000		SYSTEMS TEST EQUIPMENT
SAAAM	@	Infrared detecting test set, P/N 1978650-1
SAEEW		Transmitter, P/N 0N507630-2

3.2.4.5 Mobile Training Sets (MTS) and Resident Training Equipment (RTE). Items peculiar to the training equipment (not utilized in the operational equipment of the same type) shall be coded in the applicable operational equipment WUC manual. The abbreviation MTS or RTE, as applicable, shall be entered in parentheses following the code definition for these items. Codes for peculiar training equipment shall be restricted to items on which it is essential to collect maintenance data, such as a motor to simulate actuation or actuating devices necessary to simulate functions. Such items shall be coded following the applicable system codes for the operational equipment, with code spacing allowed to permit expansion of operational equipment codes. The following sentence shall be added to the end of paragraph 2.1 of the introduction when the preceding coding procedure is used:

> "The abbreviation (MTS or RTE, as applicable) identifies training equipment (Mobile Training Sets or Resident Training Equipment) peculiar items included in this Work Unit Code manual for recording purposes."

3.2.4.6 Site effectivity identification. Effectivity symbols may be used for identification of equipment that is not applicable to all sites (squadron or wing) of a specific Mission Design Series (MDS), missile, or space system. Numeric characters, in parentheses between the WUC and its nomenclature, shall be used for this purpose and shall be as specified by the acquiring activity (see 6.2.e). When effectivity symbols are used in accordance with this paragraph, an appropriate reference explaining their use shall be included in the introduction.

3.2.4.7 System code/end item code page entries. Paper output: Each new system WUC, except support general codes, shall begin on a new right-hand page, i.e., following completion of code assignments for 11000, Code 12000 assignments shall begin on a right-hand page.

PDF output: Each new system WUC, except support general codes, shall begin on a new page, i.e., following completion of code assignments for 11000, Code 12000 assignments shall begin on a new page.

3.2.4.8 New page entries. When WUCs for a major system or lower level assembly are carried over to a new page, the first WUC entry shall be the system code and title. Following that line entry shall be the WUC and title of the second level assembly (fourth character of the WUC) followed by " - Continued" (see figure 1). If crowding will occur the " - Continued" may be placed on the following line, i.e., assuming that the last code of the preceding page is 11AAA Aileron Actuator:

EXAMPLE:

11000	AIRFRAME	System
11AA0	OUTER PANEL - Continued	Second Level Assembly
11AAB	Aileron Follow-up Cable Guide	Third Level Assembly

3.2.4.8.1 WUC column heading. The first line entry at the top of each WUC column shall be the type of equipment to which the work unit codes apply.

EXAMPLE:

11000	AIRFRAME
25000	SOLID ROCKET
SA000	SYSTEMS TEST EQUIPMENT

3.2.4.9 Skin diagrams. Skin and access panel diagrams shall be provided for System 11 (Airframe), 14 (Flight Controls), and 23 (Nacelle skin portion). The WUCs developed and published shall provide data on the skin/access panels only with the structure beneath being reportable by other codes (see figure 3).

3.2.4.9.1 Coded segments. The view of coded segments shall be clearly distinguishable as to location on the airframe. These illustrations shall contain the following (see figure 3):

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a. Specific area. A small plan view of the aircraft with the specific area of the illustration shaded in, or a portion of the area being identified large enough to permit area identification, shall be included. A worded description, such as outboard wing, shall also be included in each diagram.

b. Particular Area. The illustration shall show the particular area of the airframe structure as a background figure, with the skin and access panel diagrams shown as exploded parts.

3.2.4.9.2 Area site. Each access panel which is assigned a number (other than a part number) shall be coded individually (see figure 3 and 6.10).

4 VERIFICATION

4.1 Verification requirements. When the technical data produced according to this specification is offered for acceptance, all tests, reviews, and verifications specified by the acquiring activity to determine that it conforms to the requirements in Section 3 of the specification, shall be accomplished as specified (see 6.2.f and 6.15).

4.2 Compliance. Technical Manuals (TMs) shall meet all requirements of Section 3 of this specification and the applicable Digital Support Suite (DSS) appendix, as specified by the acquiring activity (see 6.2). The requirements set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any requirements in this specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the government for acceptance comply with all requirements of the contract. Use of sampling inspections shall be at the discretion of the contractor, and in accordance with commercially acceptable quality assurance procedures. However, use of sampling in QA procedures does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to accept defective material.

5 PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2.g). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6 NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. WUC manuals are used to identify all "work unit" (support general and equipment identification), "type maintenance," "action taken," "when discovered," and "how malfunctioned" codes that apply to the equipment covered by the manual. Under the Air Force data collection system, each maintenance action performed on Air Force equipment must be fully and accurately documented. The volume of data collected requires the use of Automatic Data Processing (ADP) techniques. The use of ADP equipment requires that the data input be coded in such a manner that it can be read by the data processing machines. Standard codes have been developed for recording the type of maintenance being performed, the action taken to correct a deficiency, when a deficiency was discovered, and the type of malfunction that occurred. Identification codes, called "work unit codes," are used to identify the specific assembly or part within an end item on which an action was performed.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this document.

b. If the WUC manuals need not be completed prior to the start of the test program (see 3.1.7).

c. If SE is to be included in the WUC manual (see 3.2).

d. If abbreviations may be used as specified in this document (see 3.2.4.2).

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e. The numeric characters to be used in parentheses between the WUC and the nomenclature (see 3.2.4.6).

f. The requirements for test, reviews, and verifications specified for manuals developed under this specification (see 4.1).

g. Packaging requirements (see 5.1).

h. If system code manuals are to be prepared by individual series (see A.2.3.2).

i. If system code deviations authorized (see A.2.8.2).

j. If separate codes will be assigned for functionally identical (except for location) items (see A.2.8.8).

6.3 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, specifications and standards that have been authorized and assigned an Acquisition Management Systems Control (AMSC) number must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.4 Definitions. For the purpose of this document, the following definitions apply:

6.4.1 Action Taken Code (ATC). This code consists of one alphabetic or numeric character and identifies what work was done (e.g., removed, replaced, removed and reinstalled same part, etc.). The WUCs and the action taken code, collectively, identify a "unit of work" as defined in DAFI 21-101 (Maintenance Management of Aircraft) (example; Work Unit Code 23130, Igniter Assembly, Action Taken Code R - Remove and Replace) (see 3.2.2.2).

6.4.2 End item identification. End item identification is intended to aid the maintenance technicians in identifying what equipment should be entered in the "end item Work Unit Code" block of the maintenance forms (see 3.2.4.4).

6.4.3 Homogeneous group. The use of homogeneous group in this document means the use of a major sub-assembly in different end item weapon systems, i.e. ACES II seat installed in different aircraft, 'common' power generators used to supply various ground stations, etc. They often have independent managers who are responsible for the codes associated with their hardware (see 3.2.1.1).

6.4.4 How Malfunctioned Code (HMC). This code consists of three numeric characters and is used to identify how the equipment malfunctioned, (i.e., shorted, cracked, corroded, etc.) (see 3.2.2.4).

6.4.4.1 Engine related HMCs. Engine related HMCs are restricted. Engine maintenance technicians are required to use codes from the engine related group for any items which are AFMAN 20-116 (Propulsion Management For Aerial Vehicles) reportable (see 3.2.2.4.2).

6.4.5 Paper output. Specifies requirements used to develop TM data to be used as printed or paper publications. These are requirements that only apply due to the medium of the publication being paper, such as the need for volumes, binding edges, various page sizes, blank aprons for foldouts, etc.

6.4.6 PDF output. Specifies requirements used to develop TM data to be used as PDF publications. These are requirements that only apply due to the PDF medium, such as hyperlinks and bookmarks.

6.4.7 Support general codes. Support general codes are for recording production credit or repetitive tasks of a general nature (see 3.2.3 and A.2.2.1). These codes are defined in TO 00-20-2, appendix I.

6.4.8 Type Maintenance Code (TMC). This code consists of one alphabetic character and identifies the type of work performed, (e.g., scheduled or unscheduled maintenance) (see 3.2.2.1).

6.4.9 Used with, but not part of. An item which extends the use of equipment beyond its assigned functions and is issued for use with that equipment only under special circumstances, is considered as used with, but not part of, that equipment (see 3.2.4.1).

6.4.10 When Discovered Code (WDC). This code consists of one alphabetic or numeric character and identifies when a defect was discovered, or when a maintenance requirement was discovered, (i.e., during a periodic inspection, in-flight, etc.) (see 3.2.2.3).

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6.4.11 Work unit code. The WUC consists of five characters and is used to identify the system, subsystem, or component on which maintenance is required, or was accomplished. WUCs will be reviewed and approved during in-process reviews (see 3.2.4). For application and construction of WUCs, see appendix A.

6.5 Codes. Type maintenance, action taken, when discovered, and how malfunctioned codes are assigned and controlled by TO 00-20-2 (see 3.2.2).

6.6 Development and preparation. WUC, TMC, ATC, WDC, HMC, and support general WUCs will be in accordance with TO 00-20-2 (see A.2).

6.7 Test program. Collection of maintenance management type data during a test program (DAFI 99-103 or equivalent) will be in accordance with DAFI 21-101 (see 3.1.7).

6.8 Code changes or additions. Changes or additions to TMC, ATC, WDC, HMC, and support general WUCs will be in accordance with the requirements of TO 00-20-2 (see 3.2.2).

6.9 Time change, configured article, serially controlled, and warranty tracked items. These items will be approved by the acquiring activity (see 3.2.4.3).

6.10 Area site. In determining the area of skin panel size for individual coding, the rivet lines are normally used as a boundary. It is a matter of judgment to contain an area large enough to provide the ability to localize deficiencies (see 3.2.4.9.2).

6.11 Work unit code manual. Work unit code manual construction will vary depending on the concept dictated by the equipment to which it applies. SE that is commonly used in the Air Force is included in the TO -06 series work unit code manuals. Clarification of equipment applicability will be provided by the applicable system manager through the acquiring activity (see 3.2).

6.12 Page numbering examples. The first page of the TOC will be I-001 and the fifth page of the HMC engine related (alphabetical sequence) will be X-005. The first page of the support general codes will be 01-001 and the eleventh page of inertial guidance systems codes (air launched missile) will be 62-011. The first page of the engine turbine/turboprop/combustion section will be 72-40-01 and the sixth page of the landing gear/steering section will be 32-50-06 (see 3.1.4).

6.13 Acronyms. The acronyms used in this document are defined as follows:

ADP	Automatic Data Processing
AF	Air Force
AFMC	Air Force Materiel Command
AF TOPP	Air Force Technical Order Policy and Procedures
AMSC	Acquisition Management Systems Control
ASE	Aircraft Support Equipment
ATC	Action Taken Code
AWACS	Airborne Warning and Control System
CAC	Common Access Card
CDRL	Contract Data Requirements List
CD-ROM	Compact Disc Read-Only Memory
DoD	Department of Defense
DSS	Digital Support Suite
DTD	Document Type Definition
ETIMS	Enhanced Technical Information Management System
EWIS	Electrical Wiring Interconnect System
HMC	How Malfunctioned Code
IPB	Illustrated Parts Breakdown
MDS	Mission Design Series

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MTS	Mobile Training Sets
NOC	Not Otherwise Coded
NRTS	Not Repairable This Station
PCU	Pressurization Control Unit
P/N	Part Number
PDF	Portable Document Format
PKI	Public Key Infrastructure
QA	Quality Assurance
RTE	Resident Training Equipment
SE	Support Equipment
SGML	Standardized General Markup Language
SSRDI	System Subsystem Reference Designation Indicator
TCI	Time Change Item
TDT	Tag Description Table
TOC	Table of Contents
TM	Technical Manual
TMC	Type Maintenance Code
TMSS	Technical Manual Specifications & Standards
WDC	When Discovered Code
WUC	Work Unit Code

6.14 Subject term (key word listing).

Action taken code
How malfunctioned code
Support general WUCs
Type maintenance code
When discovered code

6.15 Verification requirements. (See 4.1). The Air Force Technical Order Policy and Procedures (AF TOPP) team, AFMC/A4FI, provides the specific requirements for verification of technical data developed and delivered through this specification, as well as guidance for including these requirements in the solicitation or contract (see TO 00-5-3, AF Technical Order Life Cycle Management).

6.16 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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WORK UNIT CODE	
73000	BOMBING NAVIGATION
73FC0	AN/AJN08 HEADING VERTICAL REFERENCE SET INSTALLATION - Continued
73FCC	Electronic Control and Power Supply
73FCD	Amplifier
73FCE	Amplifier Azimuth
73FCF	Compensator
73FCG	Integrator Bias
73FD0	J-4 COMPASS INSTL
73FDA	Control, Directional
73FDB	Amplifier, Servo
73FDC	Servo, Azimuth
73FE0	COMPASS AMPLIFIER INSTL
73FEA	Amplifier, Compass Signal Power, Type ME-1
73FEB	Amplifier, Compass Signal Power, Type C-1
73F99	NOC

FIGURE 1. Example of a typical work unit code page.

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TO XX-XX-06

TABLE OF CONTENTS

CODE		PAGE
	TABLE OF CONTENTS	I-001
	INTRODUCTION	II-001
	TYPE MAINTENANCE CODES	III-001
	AIRCRAFT ENGINE (SHOP TYPE MAINTENANCE CODES)	IV-001
	CLASS 1 TRAINER TYPE MAINTENANCE CODES	V-001
	ACTION TAKEN CODES	VI-001
	WHEN DISCOVERED CODES	VIII-001
	HOW MALFUNTIONED CODES AVIONIC/ELECTIRCAL/COMPUTER (ALPHABETICAL SEQUENCE)	VIII-001
	HOW MALFUNCTIONED CODES PHYSICAL/MECHANICAL (ALPHABETICAL SEQUENCE)	IX-001
	HOW MALFUNTIONED CODES ENGINE RELATED (ALPHABETICAL SEQUENCE)	X-001
	HOW MALFUNCTIONED CODES AVIONIC/ELECTRICAL/COMPUTER (NUMERICAL SEQUENCE)	XI-001
	HOW MALFUNCTIONED CODES PHYSICAL/MECHANICAL (NUMERICAL SEQUENCE)	XII-001
	HOW MALFUNCTIONED CODES NO DEFECT (NUMBERICAL SEQUENCE)	XIII-001

I-001

FIGURE 2. Example of a typical table of contents.

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TO XX-XX-06

TABLE OF CONTENTS

CODE		PAGE
	HOW MALFUNCTIONED CODES NO DEFECT (NUMBERICAL SEQUENCE)	XIV-001
	SUPPORT GENERAL CODES (EXCEPT 03 AND 04 SERIES)	XV-001
	SUPPORT GENERAL CODES	01-001
03	LOOK PHASE OF SCHEDULED INSPECTIONS	02-001
04	SPECIAL INSPECTIONS	03-001
	WORK UNIT CODE SYSTEM	
11	AIRFRAME	11-001
12	COCKPIT AND FUSELAGE COMPARTMENT	12-001
13	LANDING GEAR	13-001
14	FLIGHT CONTROL	14-001
23	TURBO FAN POWER PLANT	23-001
24	AUXILIARY POWER PLANT	24-001
41	AIR CONDITIONING PRESSURIZATION AND SURFACE ICE CONTROL	41-001
42	ELECTRICAL POWER SUPPLY	42-001
44	LIGHTING SYSTEM	44-001
45	HYDRAULIC AND PNEUMATIN POWER SUPPLY	45-001
46	FUEL SYSTEM	46-001
47	OXYGEN SYSTEM	47-001
49	MISCELLANEOUS UTILTIES	49-001
51	INSTRUMENTS	51-001
52	AUTOPILOT	52-001
55	MALFUNCTION ANALYSIS AND RECORDING EQUIPMENT	55-001
61	HF COMMUNICATIONS	61-001
62	VHF COMMUNICATIONS	62-001

I-002

FIGURE 2. Example of a typical table of contents - Continued.

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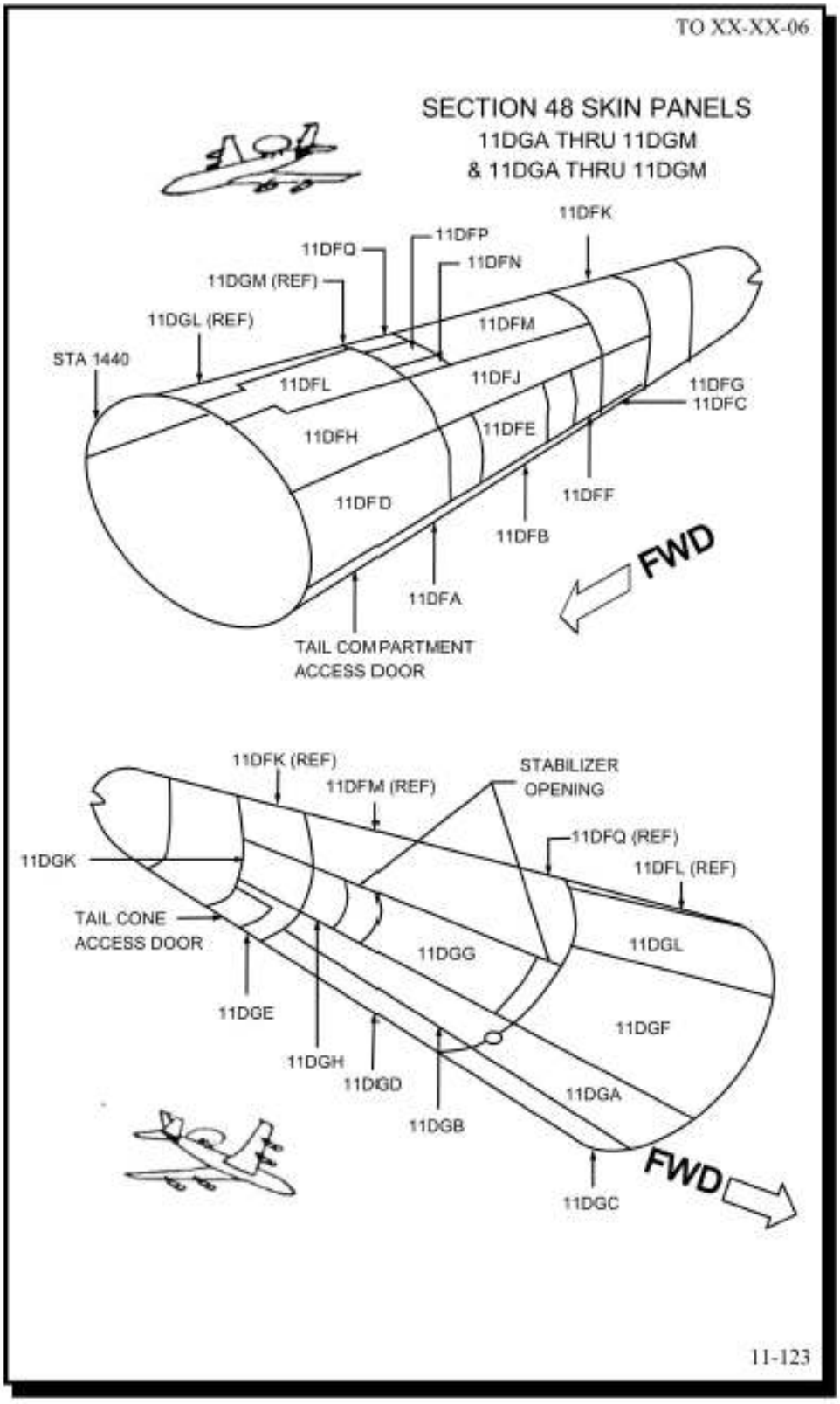


FIGURE 3. Example of a typical skin diagram.

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APPENDIX A
WORK UNIT CODE CONSTRUCTION/APPLICATION

A.1 SCOPE

A.1.1 Scope. The WUC consists of five characters and is used to identify the system, subsystem, and component on which maintenance is required or on which maintenance was accomplished. This appendix provides instructions for construction and application of WUCs. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

A.2 WORK UNIT CODE APPLICATION AND CONSTRUCTION

A.2.1 Application of work unit codes. The primary purpose of WUCs is to identify the hardware on which work has been accomplished and the relationship of hardware within a major assembly, subassembly, etc. Work unit codes will not be assigned to locations, general terms, or homogeneous group (see 6.4.3) titles. For instance, station numbers, geographic locations, and terms such as mechanical components or miscellaneous equipment may be used as a title to aid in locating WUCs within the code manual, but these titles will not have WUCs assigned. Work unit codes will not be assigned to common hardware or soft goods, such as nuts, bolts, washers, clamps, seals, packing, and O-rings. Work on these type of items is normally reported against the coded assembly on which the item is attached.

A.2.2 Work unit code - equipment identification. This is a five position code which identifies the end item, major assembly, subassembly, or component that requires maintenance. The first two positions of the WUC will be assigned in accordance with Tables I through VII, and controlled by the acquiring activity, to identify the end item of equipment. The third and fourth characters include major assemblies and subassemblies, and identify first and second levels of assembly. The fifth position of the WUC includes reparable/recoverable components, and identifies the lowest level of assembly below the end items.

A.2.2.1 Support general codes.

NOTE: Use of the assigned codes (03-/04- series only) without change is mandatory.

Support general codes are for recording production credit of repetitive tasks of a general nature and are not used for recording malfunctions, repair, Not Repairable This Station (NRTS), or condemnation actions. The multitude of tasks, which must be accomplished, precludes listing them all in the support general tables in TO 00-20-2, appendix I. This is not intended to restrict documentation of tasks. All support general work must be identified to the most appropriate code. Support general code tasks, not applicable to the equipment being coded, will not be included in the manual. Changes and/or additions must be approved by AFMC/A4FI prior to including them in TO 00-20-2, appendix I.

A.2.2.2 Scheduled and special inspections.

NOTE: Use of the assigned codes without change is mandatory.

Support general codes, for scheduled and special inspections, will be selected from TO 00-20-02, appendix I as applicable. Scheduled and special inspection codes, not applicable to the equipment being coded, will not be included in the manual. Changes and/or additions must be approved by AFMC/A4FI prior to including them in TO 00-20-2, appendix I.

A.2.3 Functional system concept. The functional system concept will be used for equipment breakout. A functional system will consist of those units which make up a system without regard to whether the units are hydraulic, electrical, pneumatic, electronic, or mechanical in nature. The components which comprise a functional system will be those components which contribute to the actual function or activation of the system. For example, the components which supply the hydraulic or pneumatic source of power will be included under System 14, “Flight Controls” which is a functional system. As another example, flight reference instruments, free air temperature, and similar instruments will be included in System 51, “Instruments”; whereas position indicators, temperature or pressure sensing, and autopilot instruments will be included under the system that they function with, such as System 52, “Autopilot.” There are two exceptions to this rule. The first is System 97, which will include all explosive

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APPENDIX A

devices regardless of the system with which they are associated. The second is System 35, which is available as an option for aircraft Electrical Wiring Interconnect System (EWIS).

A.2.3.1 Munitions work unit code breakout. Munition items of a complex nature and SE peculiar to nuclear munitions will be coded as individual end items utilizing major assembly, subassembly/mod number, and component breakout. All other munitions, components, etc., will be homogeneously grouped (see 6.4.3) and coded as major assemblies of a specific group.

A.2.3.2 Mission Design Series (MDS). Work unit code manuals will be prepared to cover each basic MDS aircraft, missile, or spacecraft. System code manuals will be prepared by individual series, if specified by the acquiring activity (see 6.2.h). This will require that the AFMC Air Logistics Complexes Single Manager ensure that a standard reporting designator is assigned for that specific mission, design, and series in accordance with TO 00-20-2.

A.2.3.3 Trainer application. Class trainers, mobile training sets, and resident training equipment will not be coded in the training equipment WUC manuals. Class III training equipment does not require detailed WUCs because one code is presently assigned for recording production credit on Class III trainers. The criteria for establishing WUCs for Class I training equipment is as follows: There must be five or more trainers in the inventory, or five or more trainers programmed for the inventory within the next fiscal year, from time of determination. They will be of sufficient cost or complexity to warrant detailed data analysis. This determination will not be predicated on cost alone.

A.2.4 Level of detail for each functional system. The WUCs for each functional system/homogeneous group (see 6.4.3) will cover each reparable item in the functional system/homogeneous group, as specified by the applicable source document (see A.2.5). Work unit codes will also be assigned to non-repairable components if they are known or suspected to be vital to successful system operation or to be significant maintenance hour consumers. These items will have a WUC assigned in order to report those critical “on-equipment” actions for reliability and maintainability evaluation purposes. In some cases it may be necessary to arbitrarily assign “interface” type items of equipment (i.e., interface connectors, tubing, couplings, wire, etc.) to a functional system. These decisions will be made after primary consideration is given to the maintenance responsibilities and training of the technicians who are required to maintain the interface items of equipment. For example, if a piece of tubing, with a regulator attached, makes the interface between the pneumatic and engine systems, and the engine technician is responsible for adjustments and maintenance of the regulator, then the tubing and associated regulator will be assigned to the engine functional subsystem.

A.2.5 Source data. Systems engineering data (i.e., engineering analysis data, equipment maintenance analysis data, and contract end item detail specifications) will be used as source data for WUC assignment. Maximum correlation between the component content of a contract end item and the assignment and grouping of WUCs is desirable. When available concurrently with the WUC manual preparation, IPB TMs will be used as source data in determining level of assembly.

A.2.6 Reserving codes for future use. The capability to ensure that new codes can be assigned for future modifications or additions of equipment will be designed into each code manual in the third, fourth, and fifth positions of the WUC. If expansion capability is designed into the original codes, new equipment can be added without disturbing the original codes. Minimum code changes are essential to prevent loss of identity for previously complied maintenance data.

A.2.7 Reuse of codes. When existing WUCs are deleted from a manual, they will not be reassigned for a minimum of ninety days following their deletion, nor without prior approval of the acquiring activity. If there is no other code or code sequence that can be used, WUCs may be reassigned by the acquiring activity without waiting ninety days.

A.2.8 Work unit code construction.

A.2.8.1 Alphabetic and numeric codes. Upper-case letters A through Z (excluding I and O) and numbers 0 through 9 will be utilized for WUC assignments as specified in subsequent paragraphs. The letters I and O will not be used in any WUC to prevent confusion with the numbers one and zero.

A.2.8.2 End item code construction. The first two characters of the WUC are alphabetic or numeric and identify the end item of equipment. These characters are assigned by Tables I thru VII and controlled by the acquiring activity (see 6.2.i). No deviations are authorized without prior approval by the acquiring activity.

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APPENDIX A

A.2.8.3 First level of assembly code construction. Alphabetic and numeric characters will be used as the third character of the WUC to designate first level of assembly below the major system. Thirty three (33) separate levels may be identified in this manner.

EXAMPLE:

33000	System
33A00 through 33Z00	First Level of Assembly
then	
33100 through 33900	First Level of Assembly

A.2.8.4 Second level assembly code construction. Alphabetic and numeric characters will be used as the fourth character of the WUC to designate second level of assembly below the major system. Thirty-three separate levels may be identified in this manner.

EXAMPLE:

33000	System
33A00	First Level of Assembly
33AA0 through 33AZ0	Second Level of Assembly
then	
33A10 through 33A90	Second Level of Assembly

A.2.8.5 Third level assembly code construction. Alphabetic and numeric characters will be used as the fifth character of the WUC to designate third or lowest level of assembly below the major system. Thirty-two levels may be identified in this manner. The number "99″ is used in the fourth and fifth positions to indicate Not Otherwise Coded (NOC).

EXAMPLE:

33000	System
33A00	First Level of Assembly
33AA0	Second Level of Assembly
33AAA through 33AAZ	Third or Lowest Level of Assembly
then	
33AA1 through 33AA8	Third or Lowest Level of Assembly
33A99	NOC (First Level of Assembly)

A.2.8.5.1 Use of "99 - NOC". The number "99″ is used in the fourth and fifth characters of the WUC, followed by "NOC." This code is used to provide a WUC for components that do not have specific codes assigned. NOC codes will relate to the first level of assembly, when the first level of assembly is the third character of the WUC. When the third character is the end item (homogeneous group see 6.4.3), the NOC code will relate to the end item rather than the first level of assembly. The "99 NOC" code will appear as the last entry under the last component in the first level of assembly (or end item for homogeneous groups) breakout. There will be a spaced separation between the last component of the first level of assembly/end item and the NOC code to provide easy recognition of the NOC code. The number "99″ will not be used in the fourth and fifth positions of the WUC to identify any specific type of equipment.

A.2.8.6 Coding capability. The construction of WUCs provides a capability to designate thirty-three first and second level assemblies (for other than homogeneously grouped items see 6.4.3), and thirty-two, third or lowest level

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APPENDIX A

assemblies. If equipment breakout exceeds these capabilities, it will be necessary to continue the listing with the next available code (note the change in the fourth character of the WUC).

EXAMPLE:

33AA0	Dual Pumping Unit
33AA8	Controller
33ABA	Controller Modulator
33A99	NOC

A.2.8.7 Homogeneously grouped code construction. Exception to the above WUC construction will be taken when utilizing homogeneous grouping (see 6.4.3) for peculiar SE. The first two characters will identify the homogeneous group, the third character will identify the end item, the fourth character will identify the first level of assembly, and the fifth character will identify the lowest level of assembly.

EXAMPLE:

CA000	Homogeneous Group
CAA00	End Item
CAAA0	First Level of Assembly
CAAAA through CAAA8	Lowest Level of Assembly
CAA99	NOC (End Item)

A.2.8.8 Multiple items serving a single system/homogeneously grouped end item. Identical components, within a system/end item performing peculiar functions, will have individual WUC assignments for each peculiar function/application. For example, a Pressurization Control Unit (PCU) in a system may contain four identical controllers, but each controller has a different usage (pressure setting) in the PCU. In this case, each controller will have a unique WUC assigned. Unless otherwise specified by the acquiring activity (see 6.2.j), items installed in multiples, within a functional system/homogeneously grouped end item (see 6.4.3), that perform within the same function/application parameters except for location will have a single code assigned. For instance, only one code will be assigned for items having left and right application, even though they have different part numbers. The acquiring activity may grant exceptions to this requirement for items that have high failure rates or warrant detailed reporting for analysis purposes. If exceptions are granted, the nomenclatures of the items will reflect part number references to distinguish between the like items. This will be done on an exception basis and will not be a common practice.

A.2.8.9 Item serving multiple systems/homogeneous grouped end items. When a single component functionally services two or more major systems/end items or lower levels of assembly, only one WUC will appear in only one of the system/homogeneous grouped end item (see 6.4.3) listings. However, the item will be listed (by identical nomenclature only) in the remaining applicable system(s)/homogeneously grouped end item(s) with a reference to the previously assigned work unit code, e.g., “Manifold Sequence Valve (Reference 33FAA).” The WUC column for such items will be left blank.

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TABLE I. Systems codes (aircraft).

10	Ground Control Station
11	Airframe
12	Cockpit and Fuselage
13	Landing Gear
14	Flight Control
15	Helicopter Rotor System
16	Escape Capsule
17	Aerial Recover System
18	Vertical Or Short Takeoff and Landing Power and Control Transmission System
19	Engine Starting
21	Reciprocating Power Plant
22	Turboprop/Turboshaft Propulsion System
23	Turbojet/Turbofan Propulsion System
(Those basic engine components as defined in TO 2J-1-24, Equipment Comprising a Complete Basic Gas Turbine Engine)	
NOTE: Constant speed drives will be coded to system 42, Electrical Power Supply	
24	Auxiliary Power Plant
25	Rocket Power Plant
26	Helicopter Rotary Wing Drive System
27	Turbojet/Turbofan Propulsion System Accessory Gear Box (B-1 only)
31	Electric Propeller
32	Hydraulic Propeller
33	Electro Hydraulic Propeller
34	Mechanical and Fixed Pitch Propellers
35	Electrical Wiring Interconnect System (EWIS)
39	Ice and Rain Protection
41	Air Conditioning, Pressurization, and Surface Ice Control
42	Electrical Power Supply
43	Electrical Multiplex
44	Lighting System
45	Hydraulic and Pneumatic Power Supply
46	Fuel System
47	Oxygen System
48	Indicating/Recording
49	Miscellaneous Utilities
51	Instruments
52	Autopilot
53	Drone Airborne Launch and Guidance Systems
54	Telemetry
55	Malfunctioned Analysis and Recording Equipment
56	Automatic All Weather Landing System
57	Integrated Guidance and Flight Control (Includes auto pilot when part of integrated system)
58	Milstar Communications System

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TABLE I. Systems codes (aircraft). - Continued

59	Crew Communications (Use if ground communications is desired, i.e., Very High Frequency, Ultra High Frequency, etc.)
60	Very Low Frequency/Low Frequency Communications
61	High Frequency Communications
62	Very High Frequency Communications
63	Ultra High Frequency Communications
64	Interphone
65	Identification, Friend or Foe
66	Emergency Communications
67	Super High Frequency/Extra High Frequency
68	Air Force Satellite Communications
69	Miscellaneous Communications Equipment
70	Nuclear Detection
71	Radio Navigation
72	Radar Navigation
73	Bombing Navigation
74	Fire Control
75	Weapon Delivery
76	Electronic Countermeasure
77	Photographic/Reconnaissance
80	Special Mission Equipment
81	Airborne Warning and Control System (AWACS)
82	Computer and Data Display (Graphic)
89	Airborne Battlefield Command Control Center (Capsule)
91	Emergency Equipment
92	Tow Target Equipment
93	Drag Chute Equipment
94	Meteorological Equipment
95	Smoke Generator, Scoring and Target Area Augmentation Systems, and Airborne Co-Operational Equipment
96	Personnel and Miscellaneous Equipment
97	Explosive Devices and Components
98	Atmospheric Research Equipment
99	R&D

TABLE II. System codes (missile), air launch.

MISSILE BASIC	
11	Airframe
13	Wing and Fin-Fold
19	Pylon
PROPULSION	
23	Gas Turbine Engine
24	Liquid Rocket
25	Solid Rocket

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TABLE II. System codes (missile), air launch. - Continued

MISSILE SUPPORT SYSTEMS	
31	Air Conditioning (including atmospheric and environmental control)
32	Pressurization (when separate from air conditioning)
33	Hydraulic Pneumatic Power Supply and Distribution
34	Electric Power Supply and Distribution
35	Electrical Distribution (Wiring Harness)
36	Component Cooling
37	Gas Driven Turbine (Mark 4 Power Plant)
39	Miscellaneous
ARMAMENT AND EXPLOSIVE DEVICES	
41	Warhead
43	Destruct
44	Arming and Fusing
45	Separation
47	Flares
FLIGHT CONTROLS	
52	Flight Controls
55	Auto Pilot
56	Flight Reference
GUIDANCE	
61	Command
62	Inertial
63	Integrated Guidance and Flight Controls
64	T.V. Guidance
65	Target Seeking, Infrared Radiation
67	Tracking (Radar)
PROPELLANT	
73	Air Breathing Engine Fuel
75	Chemical
COMMUNICATIONS AND DATA HANDLING	
91	Telemetry
92	Bomb Damage Assessment
93	Instrumentation
95	Airborne Co-Operational Equipment
96	Data Recording and Retrieval
97	Simulation
98	Reconnaissance

TABLE III. System codes, Support Equipment (SE).

AA THRU AZ	Aircraft Support Equipment (ASE), not covered under other prefixes
BA THRU BZ	Launcher
CA THRU CZ	Servicing Equipment
DA THRU DZ	Combined Servicing and Handling Equipment

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TABLE III. System codes, Support Equipment (SE). - Continued

EA THRU EZ	*
FA THRU FZ	Handling Equipment
HA THRU HZ	Environmental Control
JA THRU JZ	Electrical Generation and Distribution
KA THRU KZ	Propellant Loading and Storage
LA THRU LZ	*
MA THRU MZ	Guidance and Instrumentation
NA THRU NZ	Launch Control
PA THRU PZ	*
QA THRU QZ	Communications
RA THRU RZ	Missile Test Equipment
SA THRU SZ	Systems Test Equipment
TA THRU TZ	Training and Equipment
UA THRU UZ	Checkout Equipment
VA THRU VZ	*
WA THRU WM	Weapon System Evaluator Missile
WN THRU WZ	Mission Simulator
XA THRU XZ	*
YA THRU YZ	*
ZA THRU ZZ	Miscellaneous
* These codes are unassigned. Their utilization will require prior approval of the acquiring activity.	

TABLE IV. System codes (missile or spacecraft), ground launched.

11	Airframe/Booster Structure
12	All-Up-Round
13	Wing and Finfold
14	*
15	*
16	Orbital Craft Structure
17	Space Ferry and/or Manned Re-Entry Vehicle Structure
18	*
19	*
PROPULSION	
21	*
22	*
23	Turbo Jet
24	Liquid Rocket
25	Solid Rocket
26	Orbital Maneuvering Engine
27	*
28	Retro Rocket (excludes primary propulsion when used in retro fire mode)
29	*
MISSILE OR SPACECRAFT ENVIRONMENTAL CONTROL AND LIFE SUPPORT SYSTEMS	

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TABLE IV. System codes (missile or spacecraft), ground launched. - Continued

31	Air Conditioning (Including Atmospheric and Environmental Control)
32	Pressurization (When Separate From Air Conditioning)
33	Hydraulic/Pneumatic Power Supply and Distribution
34	Electrical Power Supply and Distribution
35	Electrical Distribution
36	*
37	Subsistence/Waste
38	Space Suit, Life Support and Personal Maneuvering Equipment
39	Miscellaneous
40	*
41	Armament and Explosive Devices
42	Initiators
43	Destruct Range Safe and Arming
45	Stage Separation
FLIGHT CONTROL	
51	Orbital Attitude Maneuvering
52	Flight Control
53	*
54	*
55	Auto Pilot
56	Flight Reference
57	Combined Controls
58	Deceleration and Surface Recovery (Excludes Retro Rocket)
59	*
GUIDANCE	
61	Command
62	Inertial
63	Integrated Guidance and Flight Controls
64	Navigator/Celestial
65	Target Seeking
66	Tracking
67	Rendezvous Radar
68	*
69	*
71	Liquid Rocket Fuel
72	Liquid Rocket Oxidizer and Hypergolic
73	Air Breathing Engine Fuel
74	Fuel and Oxidizer Pressurization Systems
75	Chemical
76	Nuclear Materials
77	*
78	*
79	*
MISSILE RE-ENTRY SYSTEM	

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TABLE IV. System codes (missile or spacecraft), ground launched. - Continued

81	Re-Entry Vehicle (Including Warhead, Arming and Fuzing)
82	Re-Entry System (Including Penetration Aids)
83	*
84	*
85	*
86	*
87	*
88	*
89	*
COMMUNICATION AND DATA HANDLING	
91	Telemetry
92	Tracking and Range Instrumentation
93	Intercom
94	Communications
95	*
96	Data Recording and Retrieval
97	*
98	Reconnaissance
99	*
* These codes are unassigned. Their utilization will require prior approval of the acquiring activity.	

TABLE V. System codes, Support Equipment/Real Property Installed Equipment (SE/RPIE).

AA THRU AZ	*
BA THRU BZ	Launcher and Launch Facility
CA THRU CZ	Servicing Equipment
DA THRU DZ	Combined Servicing and Handling Equipment
EA THRU EZ	Combined Servicing and Decontamination Equipment
FA THRU FZ	Handling Equipment
GA THRU GZ	Gas Generating Equipment
HA THRU HZ	Environmental Control
JA THRU JZ	Electrical Generation and Distribution
KA THRU KZ	Propellant Loading and Storage
MA THRU MZ	Guidance, Tracking Network and Instrumentation
NA THRU NZ	Launch Control and Launch Control Facilities
PA	Shelter Communications and Shelter Element
PB THRU PZ	*
QA THRU QZ	Communications
RA THRU RQ	Missile/Spacecraft Test Equipment
SA THRU SZ	Systems Test Equipment
TA THRU TZ	Training Equipment
UA THRU UZ	Checkout Equipment
VA THRU VZ	*
WA THRU WM	Weapon System Evaluator Missile

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TABLE V. System codes, Support Equipment/Real Property Installed Equipment (SE/RPIE). - Continued

WN THRU WZ	Mission Simulator
XA THRU XZ	Real Property Installed Equipment
YA THRU YZ	Real Property Installed Equipment
ZA THRU ZL	Miscellaneous
ZM THRU ZZ	*
* These codes are unassigned. Their utilization will require prior approval of the acquiring activity.	

TABLE VI. Systems codes, munitions.

SAMPLE HOMOGENEOUS (see 6.4.3) GROUPING IDENTIFICATION CODES	
AA	Ammunition
AAA	Shot Gun
AAAA0	12 Gauge
AAAB0	410 Gauge
AAB	Carbine, Rifle and Machine Gun
AABA0	CAL .22
AABB0	5.56 MM
AABC0	CAL .30 Carbine
AABD0	CAL .30 Rifle and Machine Gun
AABE0	7.62 MM Rifle and Machine Gun
AAC	Pistol and Revolver
AACA0	CAL .38
AACB0	CAL .45
AAD	Machine Gun
AADAO	CAL .50
AAE	Cannon
AAEA0	20 MM
AAEB0	40 MM
AAEC0	90 MM
AAF	Mortar
AAFA0	60 MM
AAFB0	81 MM
BA	Bombs
BAA	General Purpose
BAAA0	250 Pound, MK 81, Mod 1

TABLE VII. System codes, Communication Electronic (CE).

STANDARD RADAR SYSTEM CODES (Not Applicable to Navigational Radar)	
AA	Antenna System
AE	Control System
AF	Test Control System
AG	Indicator System
AK	Transmitter System

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TABLE VII. System codes, Communication Electronic (CE). - Continued

AP	Receiver System
AQ	Data Handling System
AT	Electronic Counter-Counter Measures System
AU	Timing System
AV	Radio Frequency System
AX	Tower System
BA	Communications and Inter-Communications
BC	Identification, Friend or Foe System
BE	Optics System
BF	Mapping System
BH	Support Equipment
BK	Performance Monitor
BM	Television System
BW	Miscellaneous System
BY	Multiplexer System
STANDARD COMPUTER SYSTEM CODES	
CA	Central Processing System
CF	Input/Output Control System
CL	Input/Output System
CQ	Auxiliary Storage System
CT	Power Control and Distribution System
CW	Display/Projection System
DA	Test/Monitor and Alarm System
DE	Auxiliary Devices System
DJ	Interface Systems
DM	Support Equipment
EA THRU EZ	Sites and Shelters
FA THRU FE	Imagery Intelligence Systems
FF THRU FK	Signals Intelligence Systems
FL THRU FQ	Multiple Intelligence Systems
FR THRU FV	Communications Segment
FW THRU FZ	Non-Baseline Systems
GA THRU GZ	Workstations
HA THRU HZ	Equipment Racks
JA THRU JM	Tri-Band Field Terminal
JN THRU JZ	Tri-Band Medium Earth Terminal
KA THRU KM	Modular Interoperable Surface Terminal
LA THRU LK	Servers
LL THRU LQ	Raids
LR THRU LZ	Desktop Computers
MA THRU ME	Laptop Computers
MF THRU MK	Printers
ML THRU MQK	External Computer Devices (Drives, Mice, KVM, Card Readers, Keyboard)
MR THRU MV	Router/Switch/Hub

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TABLE VII. System codes, Communication Electronic (CE). - Continued

MW THRU MZ	Monitors
NA THRU NB	Patch Panels
NC THRU NE	Fiber Optic Devices/Media Converters
FN THR NG	Multiplexers
NH THRU NK	Video Teleconference Systems/Wall Screens
NL THRU NN	Video Systems
NP THRU NQ	Audio Systems
NR THRU NS	Timing Systems
NT THRU NU	Sensor/Tap
NV THRU NW	Power Systems (Universal Power Supply, Power Distribution Unit, Generator)
NX	Telephone
NY THRU NZ	Cable
PA THRU PC	Environmental Control Unit
PD THRU PF	Antenna Systems
PG THRU PJ	Transmitters
PK THRU PM	Receivers (Assigned by CPSG CRYPTO)
QA THRU QZ	Software

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APPENDIX B
WORK UNIT CODE TECHNICAL MANUAL
MARKUP LANGUAGE TOOLS

B.1 SCOPE

B.1.1 Scope. This appendix describes the standard Air Force (AF) tagged language digital tools created for developing and delivering AF Technical Manuals (TMs). These tools are available as subsets in the Digital Support Suites (DSS) provided by the AF TMSS activity (see B.2). This appendix is a mandatory part of this detail specification. The information herein is intended for compliance.

B.1.2 Template Tool. The Document Type Definition (DTD) is the primary tool used as a template for authoring AF TMs and is based on rules outlined in MIL-PRF-28001 and ISO 8879. See B.2.1 for information about the DTD specified for this appendix subset.

B.2 DSS

The DSS is comprised of the following tools for authoring and rendering the TM. See B.3 for information about obtaining DSS component files in digital format through the TMSS activity website. For information about the current status and availability of DSS tools, see B.3.4.

B.2.1 DTD. The DTD provides the structure and content template in accordance with the content specific requirements of this specification (see 3.1). To be delivered digitally, the TM shall be tagged using the applicable DTD provided through the TMSS activity. Information concerning the tagged language type and use of DTDs currently provided, i.e., Standardized General Markup Language (SGML), may be obtained through the contacts listed under B.3.

B.2.2 Tag Description Table (TDT). The TDT provides detailed descriptions of the elements contained in the DTD. The TDT contains the element tagging structure, parent elements, full element name, source paragraph for this specification, attribute descriptions unique to the element, and entities.

B.3 OBTAINING DSS TOOLS

B.3.1 Obtaining files by users with .mil website access. The following applies to those interested in obtaining DSS component files who are on a .mil internet domain, having .mil web address access.

B.3.1.1 AF TMSS website. DTDs, TDTs, and other files in the DSS can be accessed on the TMSS website at https://techdata.wpafb.af.mil/tmss/index.html. On the web page, the “Baseline Tools” menu option in the left pane contains two bulleted options called “Specifications & Digital Support Suites (DSSs)” and “Standards & Digital Support Suites (DSSs)”. Hover the cursor over “Specifications & Digital Support Suites (DSSs)” and a listing of the TMSS specifications will appear. Hover over the desired specification and another drop down list will appear that contains an entry indicating the PDF version of the specification and other entries for the associated appendices. To obtain the preferred subset DTD, select the desired appendix from the list. The following items will appear on the downloading page: The name of the specification, the appendix number and name, the current version of the DSS, buttons to download specific DSS files provided and a “Download” button to download the entire DSS zip file.

B.3.2 Obtaining files by users with a Public Key Infrastructure (PKI) certificate or a Common Access Card (CAC). The following applies to those interested in obtaining DSS component files who have a PKI certificate or a CAC:

B.3.2.1 AF TMSS SharePoint website. DTDs, TDTs, and other files in the DSS can be accessed at the AF TMSS SharePoint website: https://usaf.dps.mil/teams/12316/default.aspx.

B.3.3 Obtaining files by users without .mil access, PKI certificate, or CAC. Those seeking to obtain DSS files who do not have .mil web access, a PKI certificate, or a CAC should contact their Government program management office or see B.3.4 to obtain information.

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APPENDIX B

B.3.4 TMSS Helpdesk assistance. Address any requests relating to the DSS by e-mail to SGMLSUPPORT@us.af.mil.

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CONCLUDING MATERIAL

Custodians:
Air Force - 16

Preparing activity:
Air Force - 16
(Project TMSS-2024-008)

Review activities:
Air Force - 01, 02, 10, 19, 70, 71, 184

NOTE: The activities listed above were interested in the document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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